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SENTINEL: AI-Powered GPU Infrastructure Management Platform

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WHY GPU CLUSTER RELIABILITY MATTERS

Modern AI and HPC workloads increasingly rely on large-scale GPU clusters for distributed training, inference, and scientific computing. However, these environments are highly dynamic and vulnerable to hardware and system-level anomalies.

KEY CHALLENGES

High Operational Complexity
Hundreds/thousands of GPUs generate large volumes of heterogeneous telemetry.

Reactive Monitoring
Traditional threshold-based monitoring often detects problems only after predefined limits are crossed.

Subtle & Early-Stage Failures
Temperature, power, utilization and communication anomalies may appear before a visible failure.

Limited Interpretability
ML-based detection can identify anomalies without clearly explaining why a GPU/node is considered abnormal.

Research Need: Move from reactive GPU monitoring toward intelligent, interpretable and proactive anomaly detection.

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RESEARCH LANDSCAPE: GPU CLUSTER RELIABILITY & ANOMALY DETECTION

EVOLUTION OF RESEARCH:

Traditional Monitoring	Statistical Anomaly Detection	Machine Learning-Based Detection	GPU Failure Prediction	Risk Ranking & Intelligent Reliability	Multi-Signal & Interpretable Monitoring
Thresholds, dashboards & rule-based alerts	Detection of deviations from normal system behaviour	Supervised & unsupervised models for anomalies and failures	Predicting failures before they affect workloads	Prioritizing high-risk GPUs/nodes using telemetry	Combining hardware, workload and observability signals

OUR LITERATURE REVIEW FOCUS:

Research Area	What We Investigate
GPU Monitoring	What telemetry is available and how it is monitored
Anomaly Detection	How abnormal GPU/node behaviour is identified
Failure Prediction	Whether failures can be predicted proactively
HPC Telemetry	How large-scale cluster behaviour is analysed

Research Area	What We Investigate
Unsupervised Learning	Detection without extensive labelled failures
Reliability & Risk Ranking	Prioritizing potentially faulty GPUs/nodes
Interpretability	Understanding why an anomaly is detected

EXISTING SYSTEMS STUDY

Existing Approach	What It Does	Major Limitation
Traditional GPU Monitoring	Monitors GPU utilization, temperature, memory, power, etc.	Mostly reactive; detects abnormal conditions after they occur
Rule/Threshold-Based Monitoring	Raises alerts when metrics cross predefined thresholds	Difficult to handle dynamic workloads and complex failure patterns
Time-Series Anomaly Detection	Detects unusual patterns in telemetry data	Often detects anomalies without understanding their impact on GPU/node health
ML-Based Fault Detection	Uses models such as VAE, LSTM and other ML techniques to identify failures	Model-specific; may lack interpretability and prioritization
HPC/Cluster Telemetry Systems	Collect large-scale CPU/GPU, node and system telemetry	Focus mainly on monitoring/analysis rather than unified predictive maintenance
AIOps Platforms	Correlate metrics, anomalies and alerts across infrastructure	Generic AIOps solutions are not specifically optimized for GPU/HPC failure behavior

Key Observation: **Existing systems can monitor and detect anomalies, but there is a gap between anomaly detection and actionable, GPU-specific predictive maintenance.**

LITERATURE REVIEW

13 research papers were studied across four major research areas

Research Area	Focus	Papers
GPU Failure & Fault Detection	GPU/node failures, faulty-machine detection, silent failures	**5**
Telemetry & Observability	GPU monitoring, HPC telemetry, interpretable observability	**3**
AIOps & Time-Series Anomaly Detection	Research Review Objectives Anomaly detection, time-series modelling, AIOps	**3**
AI-System / HPC Anomaly Analysis	AI-system anomalies and large-scale HPC anomaly detection	**2**

Research Review Objectives

- Identify existing techniques for GPU fault and anomaly detection
- Compare ML / DL methodologies used on telemetry data
- Study approaches for time-series anomaly detection
- Examine observability and interpretability
- Identify limitations in prediction, prioritization and actionable alerts
- Derive the research gap for our proposed system

Literature indicates a shift from reactive monitoring → ML-based anomaly detection → predictive and intelligent infrastructure management.

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GPU FAULT DETECTION & FAILURE IDENTIFICATION

Paper	Core Problem	Method / Approach	Key Contribution	Limitation
GPU Faulty-Machine Detection / VAE-LSTM	Detect faulty GPU machines from telemetry	VAE + LSTM	Learns temporal patterns and detects abnormal machine behavior	Primarily focuses on detection; limited failure prioritization
GPU/Node Instability, Observability & Telemetry	Identify GPU/node instability	Telemetry + observability analysis	Shows importance of monitoring low-level GPU/node signals	Monitoring-oriented; limited predictive intelligence
A Survey of Time Series Anomaly Detection in AIOps	Detect anomalies in infrastructure time series	Statistical + ML + DL approaches	Provides comprehensive view of time-series anomaly detection techniques	General AIOps focus; not GPU-specific
Interpretable GPU Cluster Monitoring – Liu et al. (2024)	Understand abnormal GPU cluster behavior	Interpretable ML / monitoring	Improves understanding of why GPU anomalies occur	Interpretation does not necessarily translate into failure prediction

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GPU FAILURE PREDICTION & PRIORITIZATION

Paper	Core Problem	Method / Approach	Key Contribution	Limitation
GPU Failures under Deep Learning – Liu et al. (2022)	Understand and identify GPU failures during DL workloads	GPU failure analysis using workload/system data	Provides insights into real-world GPU failure behavior and failure patterns	Primarily focuses on failure characterization rather than an end-to-end predictive system
Minder – Faulty Machine Detection (2022)	Detect faulty machines in large-scale computing infrastructure	ML-based faulty-machine detection	Demonstrates automated identification of machines likely to be faulty	Limited GPU-specific health prioritization and explainability
Don't Predict, Prioritize – HealthRank (2026)	Decide which infrastructure components require attention first	Health ranking / prioritization	Shifts focus from simply predicting failures to ranking system health and risk	Prioritization is not specifically designed around GPU telemetry and GPU failure modes

Key Insight: The research is evolving from "Is something anomalous?" → "Is this machine likely to fail?" → "Which machine should we prioritize?"

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GPU / HPC OBSERVABILITY & MODERN ANOMALY DETECTION

Paper	Core Problem	Method / Approach	Key Contribution	Limitation
When GPUs Fail Quietly (2026)	Detect GPU failures that do not immediately cause crashes	GPU telemetry + failure analysis	Highlights silent/degraded GPU failures that conventional monitoring can miss	Focuses primarily on identifying silent failures rather than complete predictive-maintenance workflow
Mantis – HPC Telemetry (2026)	Monitor large-scale HPC systems using telemetry	Large-scale HPC telemetry analysis	Demonstrates the importance of continuous, high-volume system telemetry	Broad HPC focus; not specifically designed for GPU health ranking
Reveal – AI-System Anomalies (2025)	Detect anomalies in complex AI systems	AI-system observability and anomaly analysis	Shows how anomalies can emerge across modern AI infrastructure	Broader AI-system scope; does not specifically target GPU failure prediction

AIOPS & TIME-SERIES ANOMALY DETECTION

Paper	Core Problem	Method / Approach	Key Contribution	Limitation
AI-Based Anomaly Detection in Time Series – Ahmed et al. (2021)	Detect anomalies in complex time-series data	AI/ML-based time-series anomaly detection	Provides foundational techniques for identifying abnormal temporal patterns	General-purpose; not tailored to GPU/HPC telemetry
AIOps Time-Series Anomaly Detection – Zhong et al. (2023)	Detect infrastructure anomalies in AIOps environments	Time-series anomaly detection + AIOps	Demonstrates how temporal analysis can support intelligent IT operations	Does not specifically address GPU failure modes or GPU health ranking
Anomaly Detection in HPC – Lu et al. (2026)	Detect anomalies in large-scale HPC systems	HPC-specific anomaly detection	Brings anomaly detection closer to large-scale computing environments	Detection-focused; limited end-to-end predictive maintenance

Key Takeaway:
AIOps research provides strong anomaly-detection methodologies, but GPU-specific predictive maintenance requires domain-aware telemetry analysis.

AIOPS & TIME-SERIES ANOMALY DETECTION

Paper	GPU-Specific	Time-Series	Anomaly Detection	Failure Prediction	Explainability	Prioritization
GPU Faulty-Machine / VAE-LSTM	✓	✓	✓	✓	–	–
GPU/Node Instability & Telemetry	✓	✓	✓	–	–	–
AIOps Time-Series Survey	–	✓	✓	–	–	–
Interpretable GPU Monitoring	✓	✓	✓	–	✓	–
GPU Failures under Deep Learning	✓	✓	–	–	✓	–
Minder	–	✓	✓	✓	–	–
HealthRank	–	✓	–	✓	✓	✓
When GPUs Fail Quietly	✓	✓	✓	✓	–	–
Mantis	–	✓	✓	–	–	–
Reveal	–	✓	✓	–	✓	–
Ahmed et al.	–	✓	✓	–	–	–
Zhong et al.	–	✓	✓	–	–	–
Lu et al.	–	✓	✓	–	–	–

No single existing approach combines GPU-specific telemetry, temporal anomaly detection, failure prediction, explainability and health-based prioritization into one integrated system.

SOCIETAL & ENVIRONMENTAL RELEVANCE

🌱 Environmental Impact

Reduces unnecessary GPU resource consumption by identifying inefficient or degrading GPUs early.
Helps prevent repeated computation caused by GPU failures and interrupted workloads.
Supports more efficient utilization of energy-intensive AI/HPC infrastructure.
Contributes to sustainable and energy-efficient computing.

⚙️ Operational & Economic Impact

Early detection → reduced downtime for GPU clusters.
Helps system administrators prioritize high-risk GPUs/nodes instead of investigating every alert.
Reduces potential maintenance costs and resource wastage.
Improves reliability of infrastructure supporting AI and data-intensive workloads.

👥 Societal Relevance

More reliable AI infrastructure benefits applications such as:
Healthcare & medical AI
Scientific research
Climate and weather modelling
Autonomous systems
Large-scale data analytics
Supports the responsible and efficient use of increasingly resource-intensive AI systems.

PROBLEM STATEMENT

Modern GPU clusters generate large volumes of telemetry such as:
GPU Utilization • Temperature • Memory • Power • Performance • Error Metrics

However, existing monitoring approaches often rely on static thresholds or isolated anomaly detection, making it difficult to identify gradual degradation and impending failures early.

The Problem

There is a need for an intelligent GPU cluster observability system that can continuously analyse telemetry, detect abnormal behaviour, estimate GPU/node health and failure risk, and prioritize critical components before they cause significant performance degradation or downtime.

Current Challenges

🔴 Reactive monitoring – problems are often identified after degradation or failure.
🔴 High telemetry volume – difficult for operators to manually analyse continuously.
🔴 Complex temporal patterns – failures may develop gradually rather than through a single threshold violation.
🔴 Alert overload – multiple anomalies may occur simultaneously without clear priority.
🔴 Limited interpretability – anomaly alerts may not clearly explain the underlying issue.
🔴 Fragmented solutions – monitoring, anomaly detection, prediction and prioritization are often handled separately.

OBJECTIVES

Primary Objective

To develop SENTINEL, an intelligent GPU cluster observability platform that uses telemetry and AI/ML-based analysis to detect anomalies, assess GPU/node health, identify potential failures early, and support predictive maintenance.

Specific Objectives

01 – Continuous Monitoring
Collect and monitor relevant GPU and node telemetry continuously.
02 – Anomaly Detection
Identify abnormal patterns in GPU cluster telemetry using time-series AI/ML techniques.
03 – Health Assessment
Evaluate the health and operational condition of individual GPUs and nodes.
04 – Failure-Risk Identification
Detect patterns indicating potential GPU/node degradation or failure before critical breakdown.
05 – Risk Prioritization
Rank GPUs/nodes according to their health or failure risk, enabling operators to focus on the most critical components.
06 – Actionable Observability
Present detected issues through an interpretable and actionable monitoring interface.

REQUIREMENTS & FEATURES

REQUIREMENT	DESCRIPTION
Telemetry Collection	Collect GPU and node-level metrics continuously
Data Processing	Clean, normalize and prepare telemetry for analysis
Health Assessment	Generate a health/risk status for GPUs and nodes
Failure-Risk Analysis	Identify components showing signs of potential failure

NON-FUNCTIONAL REQUIREMENTS

Scalability – support multiple GPUs/nodes and increasing telemetry volume

Reliability – provide consistent monitoring and detection

Low Latency – process incoming telemetry with minimal delay

Interpretability – provide understandable reasons/evidence for detected issues

Maintainability – modular architecture for future model/feature additions

SCOPE

GPU & Node Monitoring

- Continuous collection of GPU and node-level telemetry
- Monitoring of metrics such as utilization, temperature, memory, power and performance

Intelligent Analysis

- Preprocessing and analysis of telemetry as time-series data
- AI/ML-based anomaly detection
- Identification of abnormal/degrading behaviour

Health & Risk Assessment

- GPU/node health evaluation
- Failure-risk identification
- Risk-based prioritization of critical components

Observability

- Centralized dashboard for cluster health
- Visualization of telemetry, anomalies and risk statuses

Telemetry → Analysis → Detection → Health/Risk Assessment → Prioritization → Alert

SENTINEL focuses on intelligent observability and predictive maintenance support—not physical intervention.

METHODOLOGIES / TECHNIQUES CONSIDERED

Based on the literature review, we considered approaches across data processing, anomaly detection, prediction and prioritization.

METHODOLOGY SELECTION CRITERIA

The techniques were evaluated based on:

GPU/HPC relevance
Ability to handle time-series telemetry
Anomaly-detection capability
Ability to capture temporal behaviour
Computational feasibility
Interpretability
Suitability for predictive maintenance
Ability to support health/risk prioritization

Overall Methodological Direction

SENTINEL combines time-series telemetry analysis with AI/ML-based anomaly detection, health assessment and risk prioritization.

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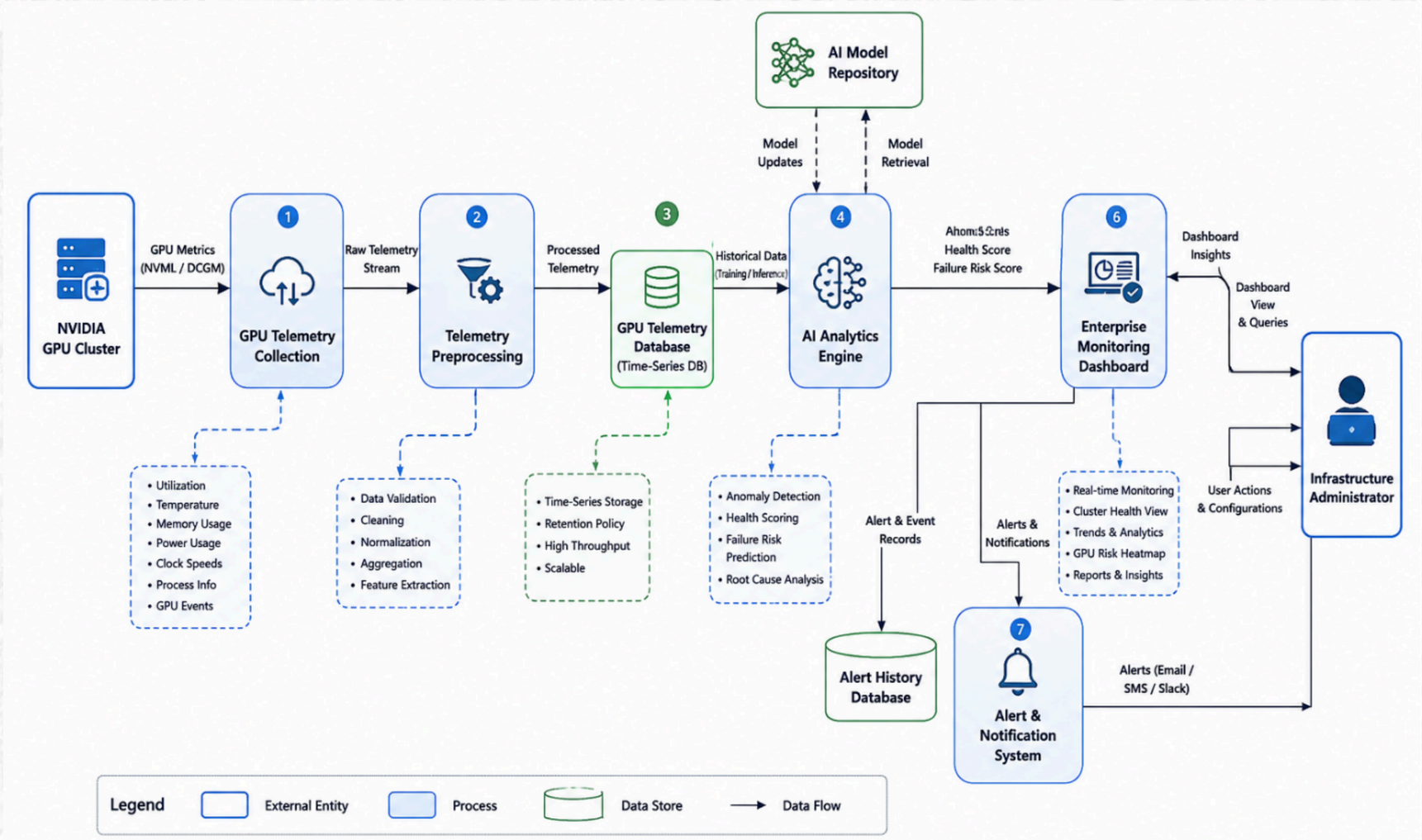
PROPOSED ARCHITECTURE



HARDWARE & SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS

- Processor: Intel Core i5 (10th Generation or above) / AMD Ryzen 5 or equivalent
- RAM: Minimum 16 GB (32 GB Recommended)
- Storage: Minimum 512 GB SSD
- GPU: NVIDIA GPU with CUDA support (GTX/RTX Series or Data Center GPU preferred for telemetry collection)
- Internet Connectivity: Stable broadband connection for software installation, updates, and cloud integration

SOFTWARE REQUIREMENTS

- Operating System: Windows 11 / Ubuntu 22.04 LTS (Preferred)
- Programming Language: Python 3.11+, TypeScript, JavaScript
- Backend Framework: FastAPI
- Frontend Framework: React.js
- UI Framework: Tailwind CSS
- Database: PostgreSQL with TimescaleDB Extension
- Machine Learning Libraries: Scikit-learn, PyTorch
- GPU Monitoring Libraries: NVIDIA Management Library (NVML), NVIDIA Data Center GPU Manager (DCGM)
- Data Visualization: Recharts / Chart.js
- API Testing: Postman
- Version Control: Git & GitHub
- Development Environment: Visual Studio Code
- Containerization: Docker & Docker Compose (Optional)

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